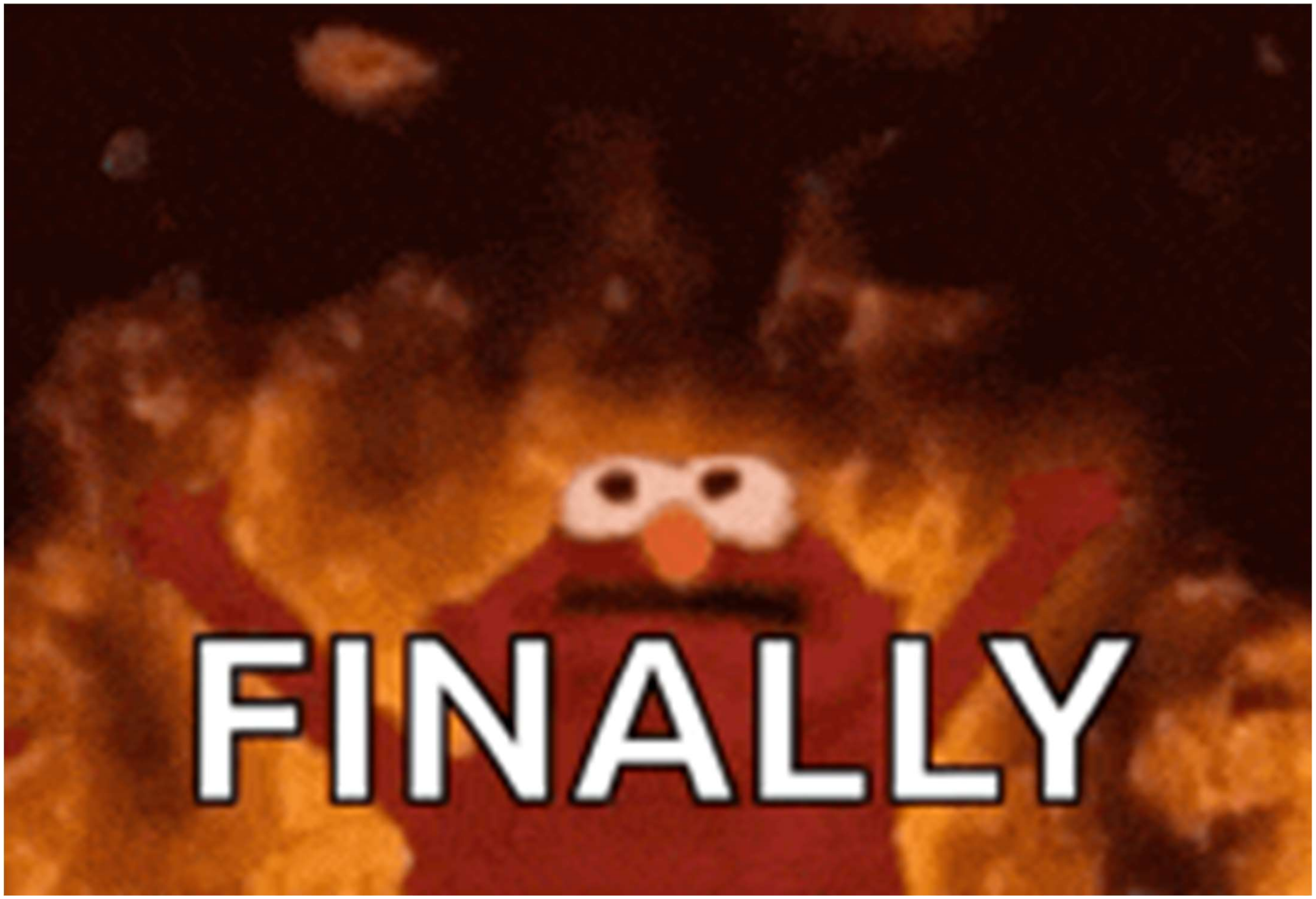
Course Introduction

SE-350/SE-450, Spring-2026
Artificial Intelligence (AI)

Dr. Kanwal Yousaf
Department of Software Engineering
UET Taxila
e-mail: kanwal.yousaf@uettaxila.edu.pk

Some slides material from
Emma Pierson, University of California, Berkeley


FINALLY

AI Community

- Hot area right now…!!
- Important course of Computing in various universities
- Specializations & trainings
- Projects funded by organizations or governments maybe
- Lastly, its everywhere now….

Course required background

- No strict pre-reqs enforced as such, however
- It is good to have:
  - programming fundamentals (Python will be used throughout the course)
  - Familiarity with probability, discrete mathematics, and linear algebra.
  - Willingness to learn

This course is not about memorizing equations. It's about learning how machines think and how we teach them to learn

Course learning outcomes

- At the end of this course, students will be able to
  - Describe the core concepts and approaches of AI (C2, PLO-1)
  - Analyze AI techniques for real-world problem-solving and decision making (C4, PLO-4)
  - Develop AI models using appropriate methods for various applications (C6, PLO-3)

*BT Level=Bloom's Taxonomy Level

C(Cognitive Domain): C1(Remembering), C2(Understanding), C3(Applying), C4(Analyzing), C5(Evaluation), C6(Creating)

Group to join for updates

- All course material and important information will be shared in the MS Teams group below:

Course Material

- Recommended books
  - Artificial Intelligence: A Modern Approach (4th Edition), Stuart Russell and Peter Norvig. Publisher: Pearson Education Ltd, 2022.
  - The AI-Driven Leader: Harnessing AI to Make Faster, Smarter Decisions, Geoff Woods. Publisher: AI Thought Leadership™, 2025.
- Reference books
  - Introduction to Machine Learning (4th Edition), Ethem Alpaydın. Publisher: The MIT Press, 2020.
- and, online sources

What will you learn from this course?

- The foundations of the effective techniques used in AI
  - Basics first: intelligent agents, problem formulation, state-space search, and classical AI algorithms for solving well-defined problems.
  - From foundations to modern systems: how deep learning, generative models, large language models, and reinforcement learning come together in today's AI applications.
- A big-picture view of how AI systems are built, evaluated, and deployed responsibly, including ethics, explainability, bias, and societal impact.

What will you learn from this course? (In depth)

Lecture #	Lecture Contents	Task
1.	**Introduction to Artificial Intelligence** • What is AI? Myths vs reality • History and evolution of AI • Intelligent agents and types	
2.	**Problem Formulation & State Space Search** • Problem-solving agents • State space representation • Initial state, actions, transition model, goal test, path cost	
3.	**Uninformed Search Algorithms** • Breadth First Search (BFS) • Depth First Search (DFS) • Depth-Limited Search • Iterative Deepening Search (IDS) • Uniform Cost Search • Time and space complexity	
4.	**Informed Search Algorithms** • Heuristics: admissible and consistent • Greedy Best-First Search • A* Search • Comparison of search strategies	**Assignment 01 (Due)**
5.	**Local Search and Optimization** • Hill Climbing • Simulated Annealing • Genetic Algorithms (GA)	
6.	**Adversarial Search and Game Playing** • Game environments • Minimax algorithm • Alpha-Beta pruning • Evaluation functions	
7.	**Constraint Satisfaction Problems (CSPs)** • Variables, domains, constraints • Backtracking search • Forward checking • Constraint propagation • Real-world CSP examples (timetabling, Sudoku)	**Quiz 01**
8.	**Knowledge Representation** • Importance of knowledge representation • Propositional Logic • First-Order Logic (FOL) • Syntax and semantics • Representing facts and rules	

What will you learn from this course? (In depth) (cont...)

Lecture #	Lecture Contents	Task
9.	**Inference and Reasoning** • Inference in propositional logic • Forward chaining • Backward chaining • Knowledge base construction • Rule-based expert systems	
10.	**Handling Uncertainty** • Why uncertainty exists in AI • Probability basics for AI • Bayesian Networks • Markov Decision Processes (MDPs)	**Quiz 02**
11.	**Introduction to Machine Learning** • What is learning? • Supervised vs unsupervised learning • Training, testing, overfitting • Basic ML algorithms overview (Naïve Bayes, k-NN)	**Assignment 02 (Announced)**
12.	**Neural Networks** • Introduction to Neural Networks • Perceptron • Activation Functions, Backpropagation, Loss Functions	

Lecture #	Lecture Contents	Task
13.	**Neural Networks** • Single-layer vs. Multi-layer neural networks • Training neural networks • Introduction to Deep Learning	
14.	**Reinforcement Learning** • Learning from interaction • Agent, environment, reward • Q-learning	
15.	**Modern AI Systems and Trends** • Computer Vision and NLP applications (LLMs) • Generative AI (text, image, video)	**Assignment 02 (Due)**
16.	**Ethics, Societal Impact, and Future of AI** • Bias and fairness • Explainable AI (XAI) • AI safety and misuse	

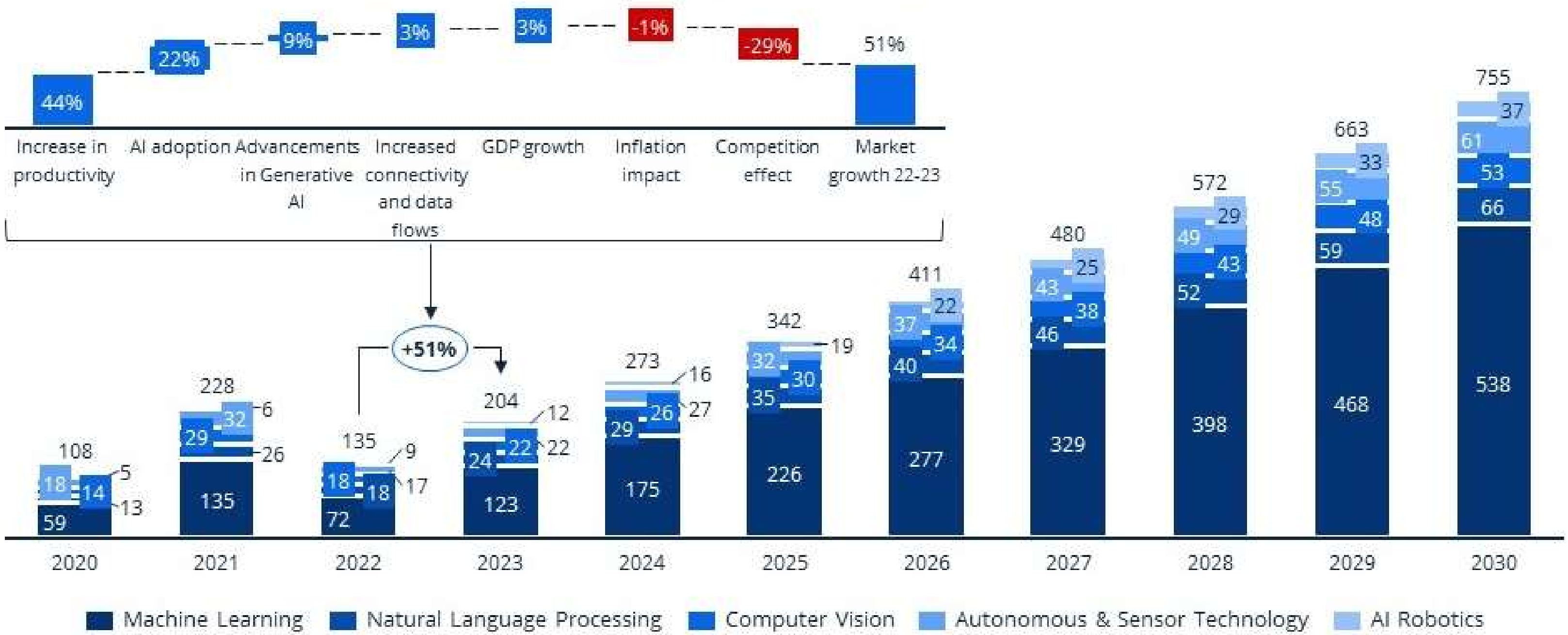
Grading criteria

S. No.	Assessment Items	No. of items	%age
1	Assignments (Theoretical + Practical)	2	5+**15**
2	Quizzes	2	5
3	**Presentation/ Demonstration/Viva**	**--**	**--**
4	Mid Exam (after 8 weeks)	1	25
5	Final examination (after 16 weeks)	1	50
	Total	**--**	**100**


AI's Journey Through the Ages
UNIVERSITY OF ENGINEERING AND TECHNOLOGY
TAXILA
Turing Test
Early Concepts
1980s–1990s
Symbolic AI
Machine Learning
Machine Learning
Expert Systems
Neural Networks
Big Data &
Deep Learning
Big Data
Deep Learning
AI Revolution
Robotics
Computer Vision
NLP & LLMs
Future
Advanced AI

AI market

Global AI market value development in billion US$ and YoY growth rate explanation for 2022-2023



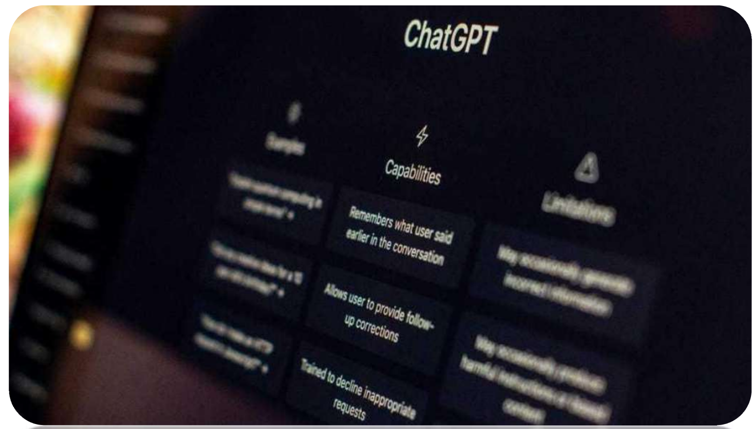
Tech markets CAGR 2020-2027

AI	**23.7%**
IoT	20.5%
Cloud	18.6%
Cybersecurity	9.9%
Semiconductors	8.0%
IT Services	7.8%
Robotics	6.2%
Software	6.1%

Source: https://www.statista.com/site/insights-compass-ai-ai-market-overview

AI is having real-world impact

- **Public imagination**
  - Text assistants
  - Image generation

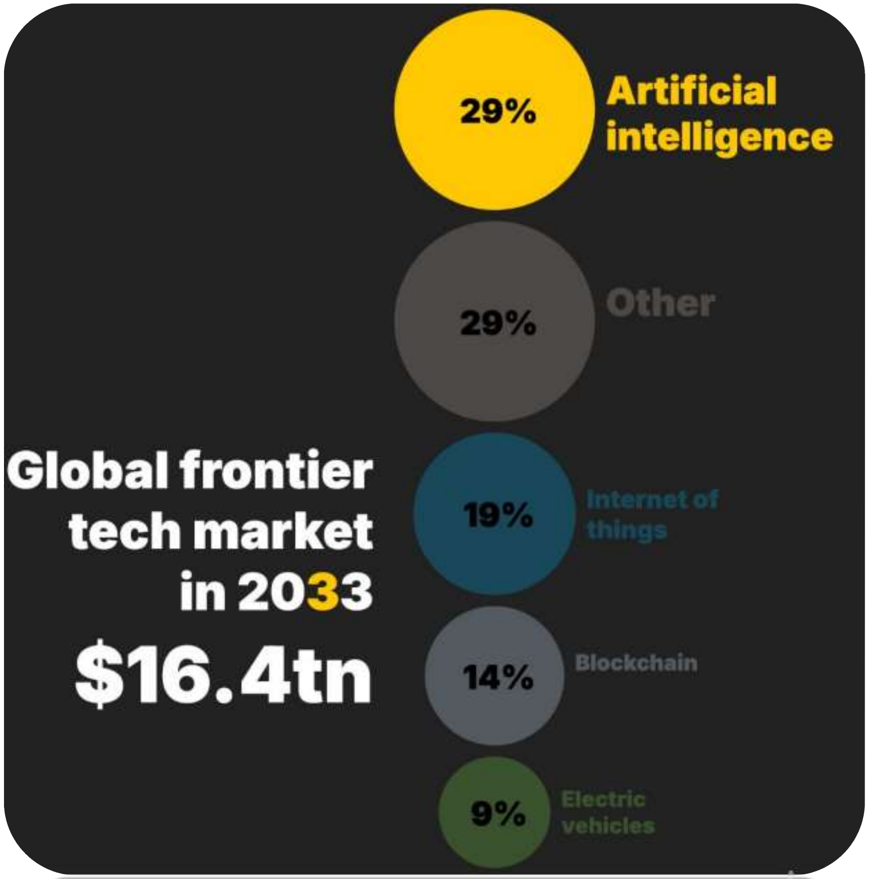


an espresso machine that makes coffee from human souls, artstation | panda mad scientist mixing sparkling chemicals, artstation | a corgi's head depicted as an explosion of a nebula | vibrant portrait painting of Salvador Dalí with a robotic half face | a shiba inu wearing a beret and black turtleneck

AI is having real-world impact (cont…)

- **Economy**
  - $4.8 trillion by 2033

    (25x increase in a decade)



https://unctad.org/publication/technology-and-innovation-report-2025

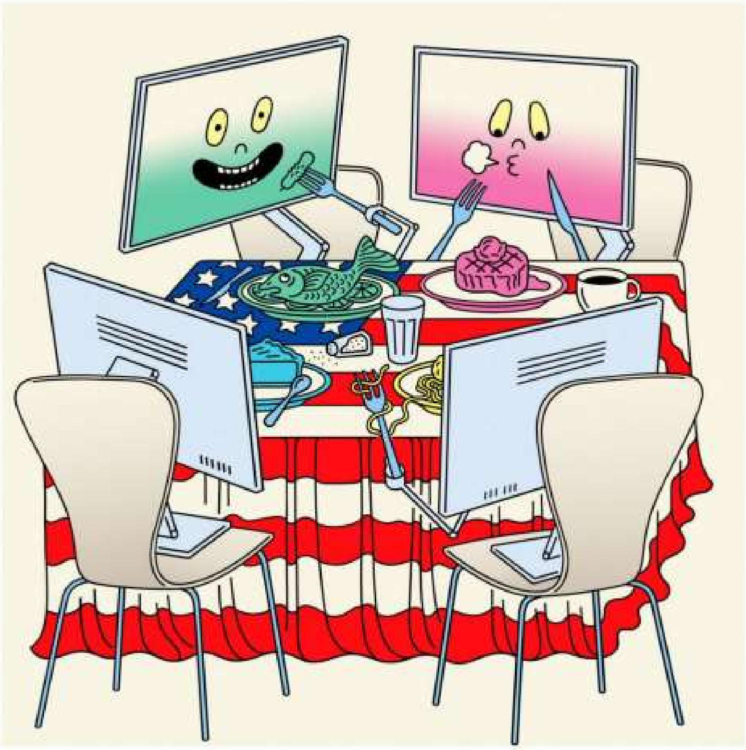
AI is having real-world impact (cont…)

- **Politics**

The New York Times

OPINION

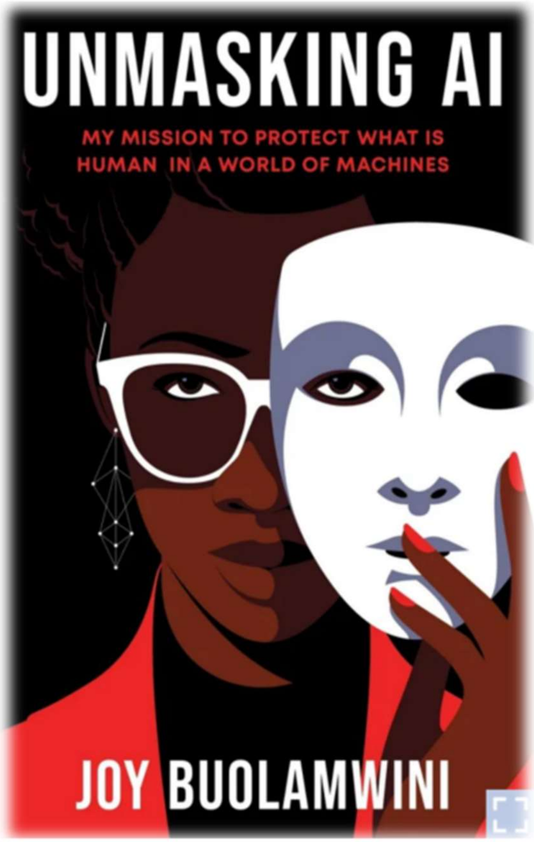
How A.I. Chatbots Become Political

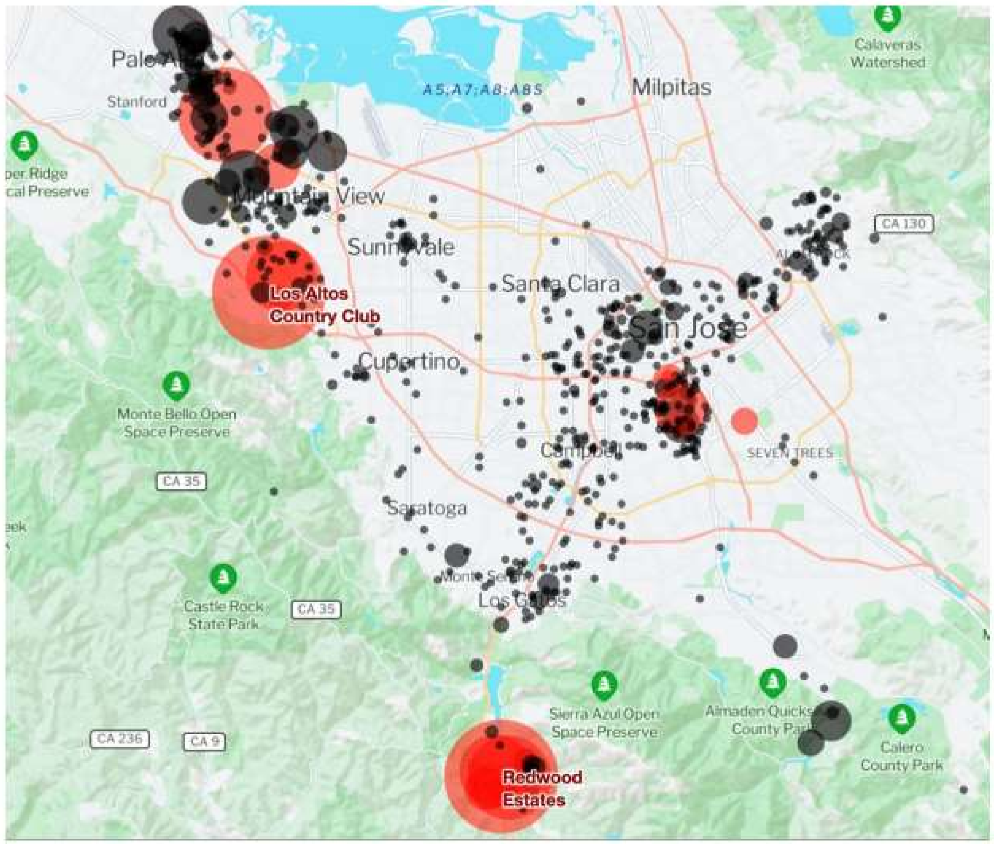


https://www.nytimes.com/2025/02/11/world/europe/vance-speech-paris-ai-summit.html

AI is having real-world impact (cont…)

- **Law**





AI for Scaling Legal Reform: Mapping and Redacting Racial Covenants in Santa Clara County. Surani et al, 2024.

AI is having real-world impact (cont…)

- **Law**

Legal Document Summarization Using Natural Language Processing

Authors

Ishaa Arshad 21-SE-08
Izza Shahzad 21-SE-18
Madiha Shah 21-SE-44

Supervisor

Dr. Kanwal Yousaf

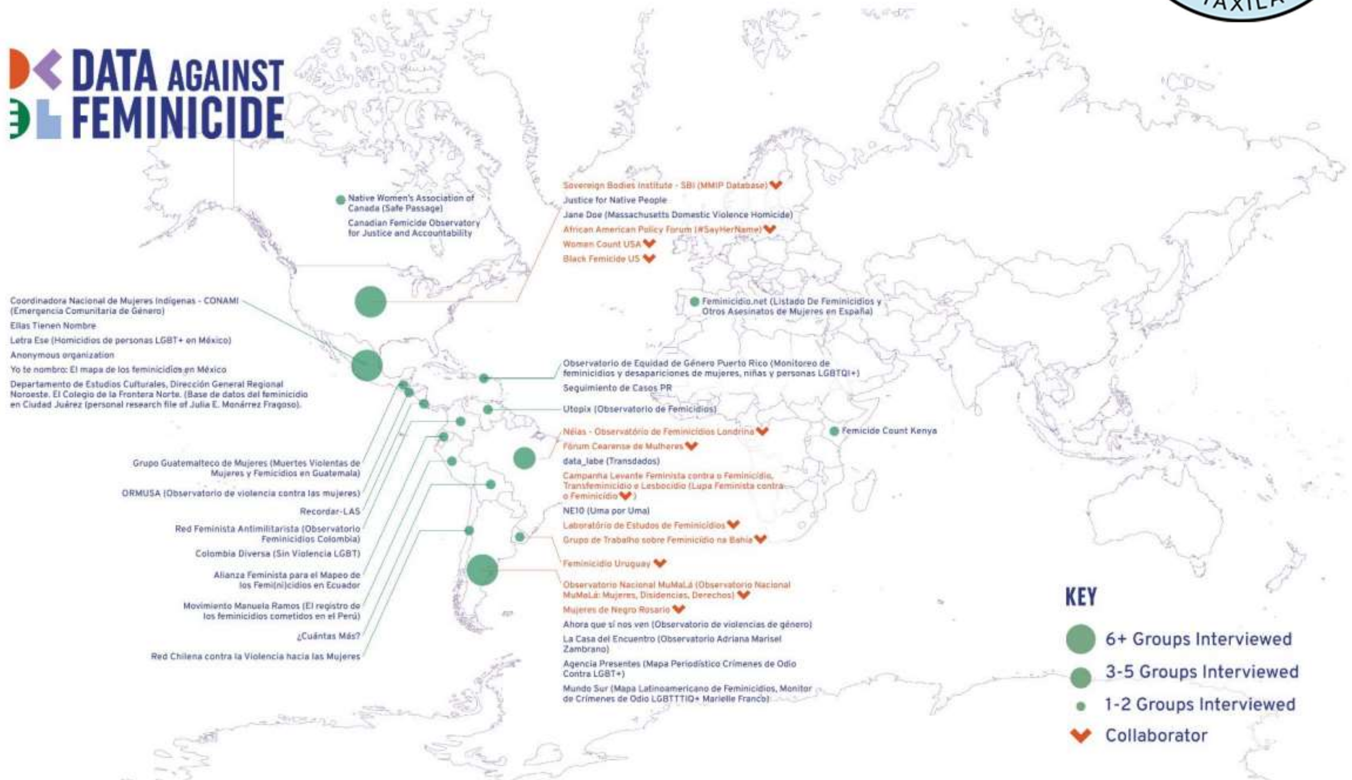
Lecturer

DEPARTMENT OF SOFTWARE ENGINEERING

FACULTY OF TELECOMMUNICATION AND INFORMATION ENGINEERING

UNIVERSITY OF ENGINEERING AND TECHNOLOGY TAXILA

JUNE 2025



https://dataplusfeminism.mit.edu/

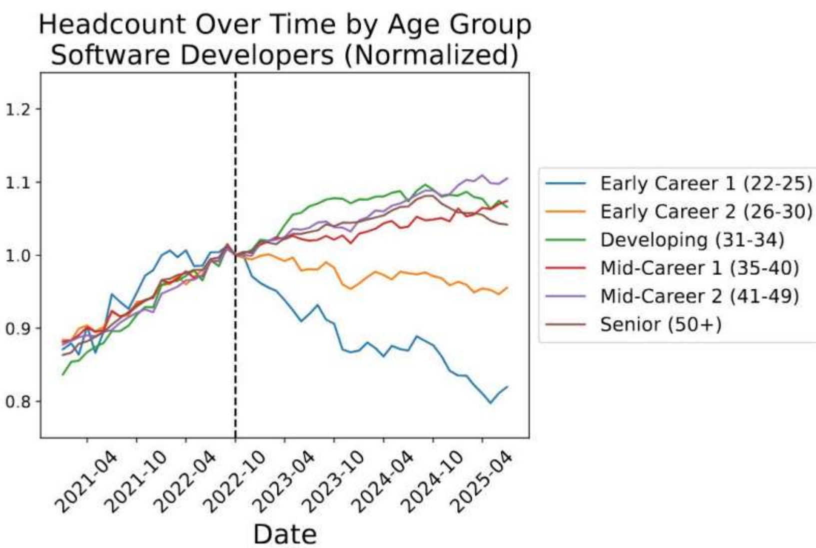
AI is having real-world impact (cont…)

- **Labor**

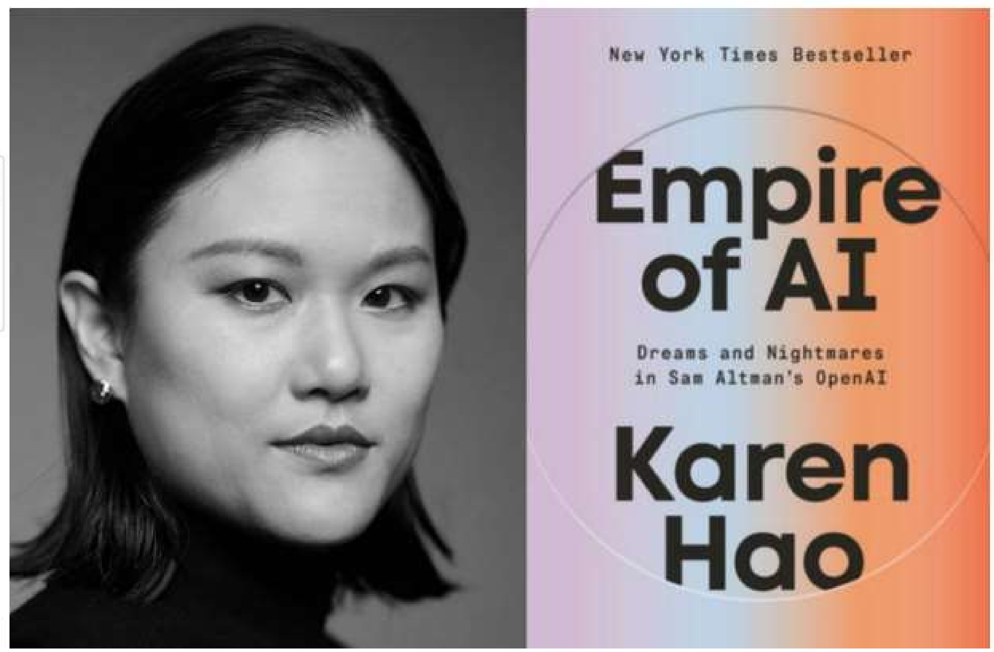
The human labor behind AI chatbots and other smart tools

Data labeling is an important step in developing artificial intelligence but also exposes the people doing the work to harmful content.

MarketWatch, 2023



Stanford Digital Economy Lab, 2025



AI is having real-world impact (cont…)

- **Science**
  - Chemistry, Physics & Biology

Science & technology | The 2024 Nobel prizes

AI wins big at the Nobels

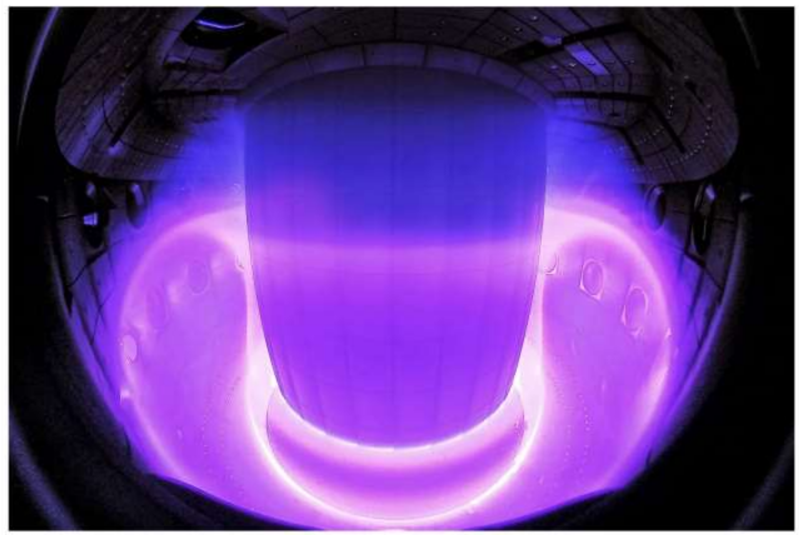
Awards went to the discoverers of micro-RNA, pioneers of artificial-intelligence models and those using them for protein-structure prediction

The Economist, 2024

AMIT KATWALA SCIENCE FEB 16, 2022 11:00 AM

DeepMind Has Trained an AI to Control Nuclear Fusion

The Google-backed firm taught a reinforcement learning algorithm to control the fiery plasma inside a tokamak nuclear fusion reactor.



PHOTOGRAPH: CURDIN WÜTHRICH, SPC/EPFL

Wired, 2022

AI is having real-world impact (cont…)

- **Science**
  - Conservation

Nature Communications, 2022

AI is having real-world impact (cont…)

- **Education**

BREAKING

ChatGPT In Schools: Here's Where It's Banned—And How It Could Potentially Help Students

Arianna Johnson Forbes Staff

I cover the latest trends in science, tech and healthcare.

Follow

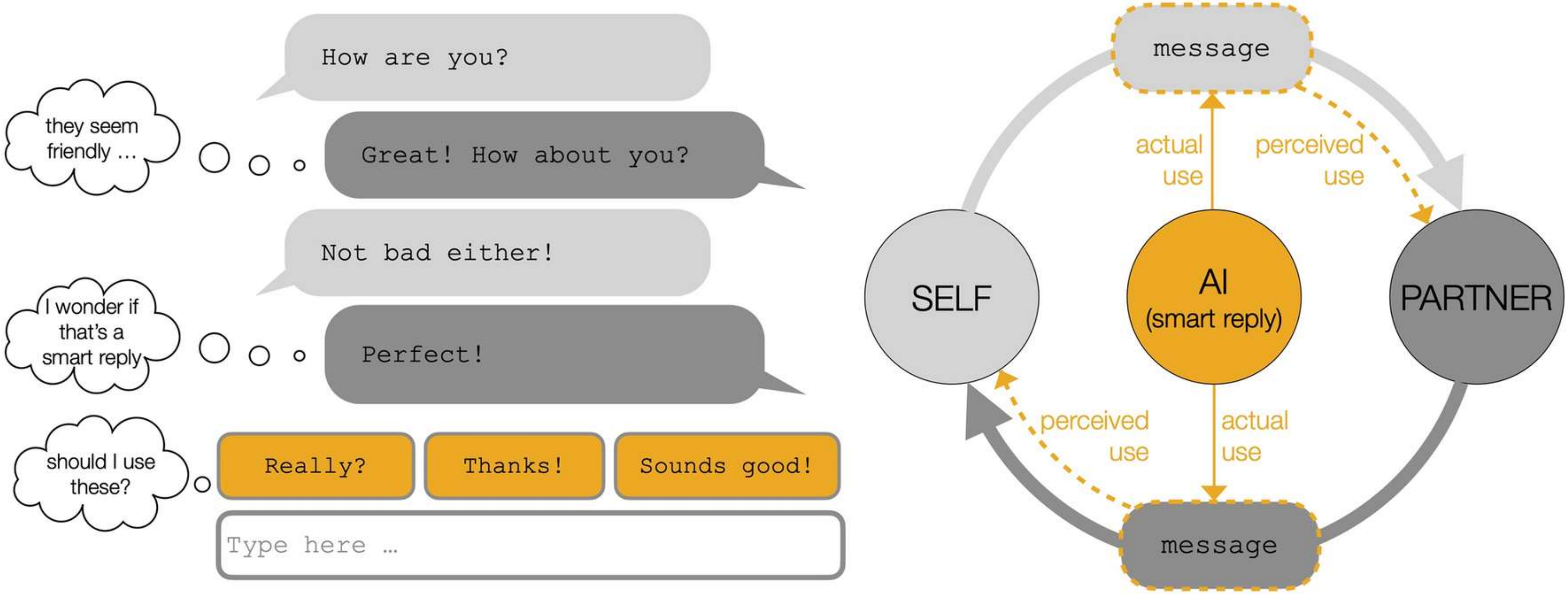
2

Jan 18, 2023, 02:31pm EST

Forbes, 2023

AI is having real-world impact (cont…)

- **Social Interaction**

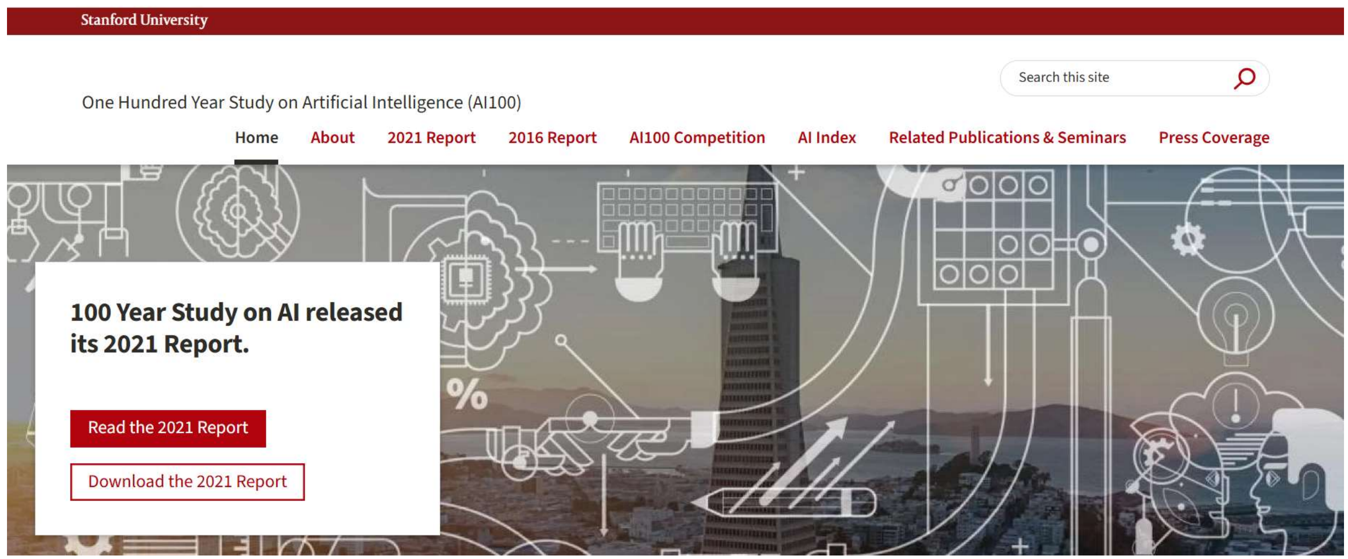


https://www.nature.com/articles/s41598-023-30938-9

Ok, but what actually is AI???

Definition of Artificial Intelligence

- "Artificial Intelligence (AI) is a science and a set of computational technologies that are inspired by — but typically operate quite differently from — the ways people use their nervous systems and bodies to sense, learn, reason, and take action."



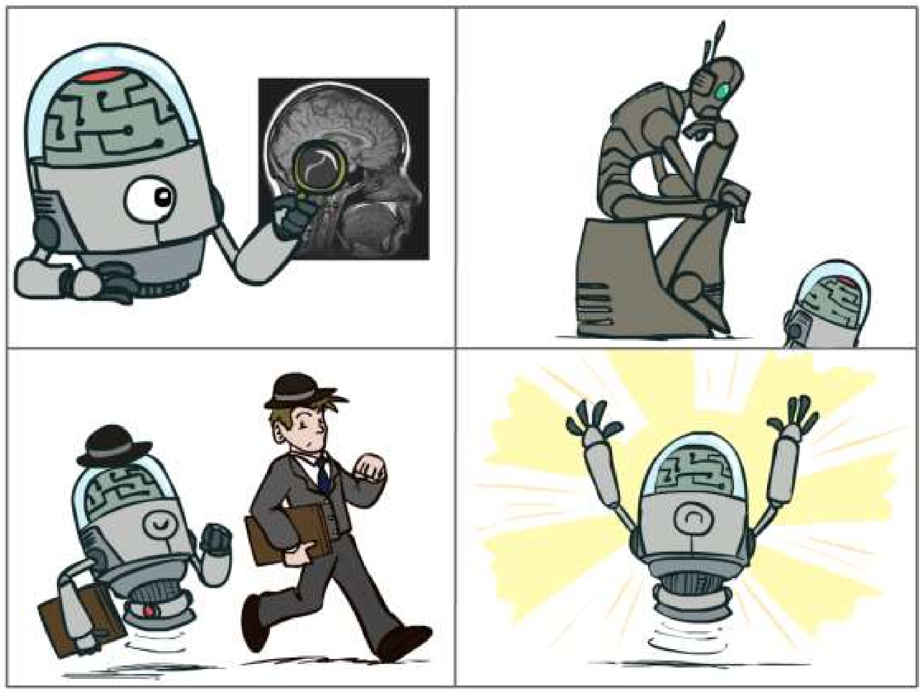
https://ai100.stanford.edu/sites/g/files/sbiybj18871/files/media/file/ai100report10032016fnl_singles.pdf

In simpler terms…

- The science of making machines that:

Think like people

Think rationally



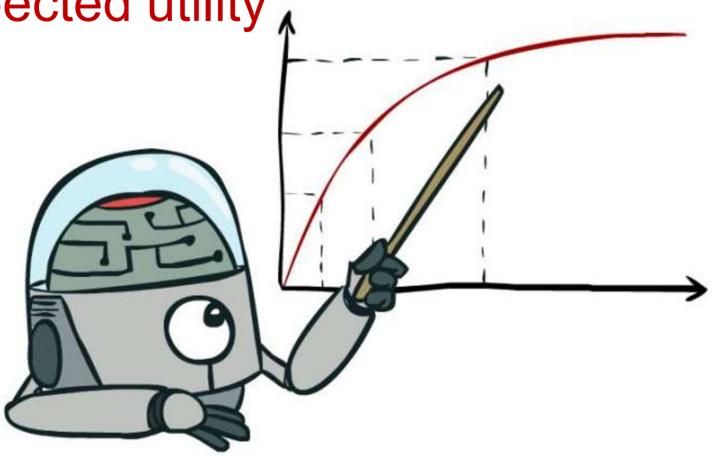
Act like people

Act rationally

Rational Decisions

- Should we make machines that do rational decisions…
- **Rational**: *maximally* achieving pre-defined goals
- Goals are expressed in terms of the **utility** of outcomes
- World is uncertain, so we'll use **expected** utility
- Being rational means acting to maximize your expected utility

A better title for this course might be:
Introduction to Computational Rationality



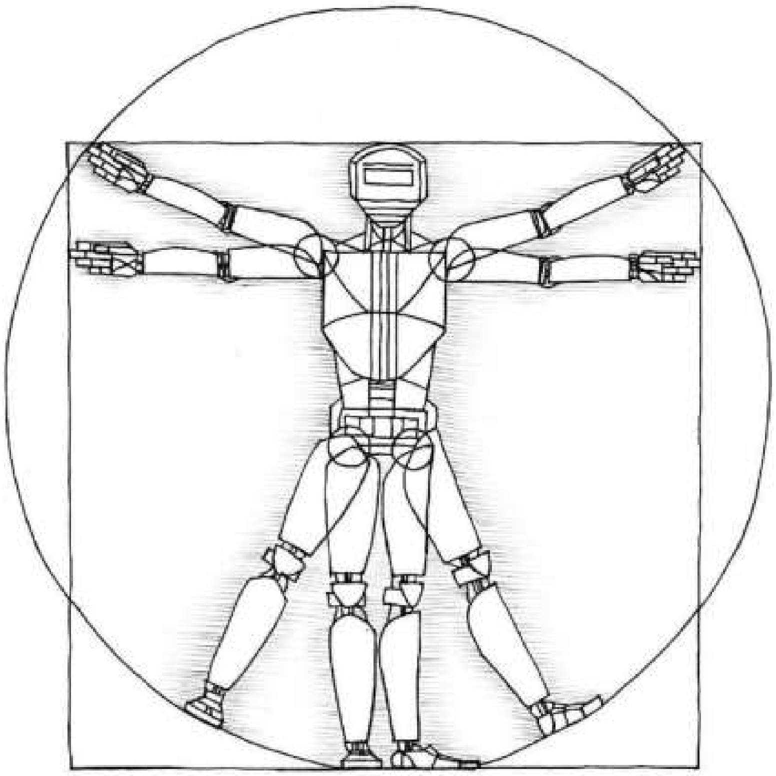
Perspectives on Intelligence

Skills-based perspective

- "A system is only intelligent if it can do [X]."
  - Play chess?
  - Learn from experience?
  - Use words properly?
  - Make mistakes?
  - Not make mistakes?

Perspectives on Intelligence (cont…)

Embodiment perspective (Rodney Brooks)



Perspectives on Intelligence (cont…)

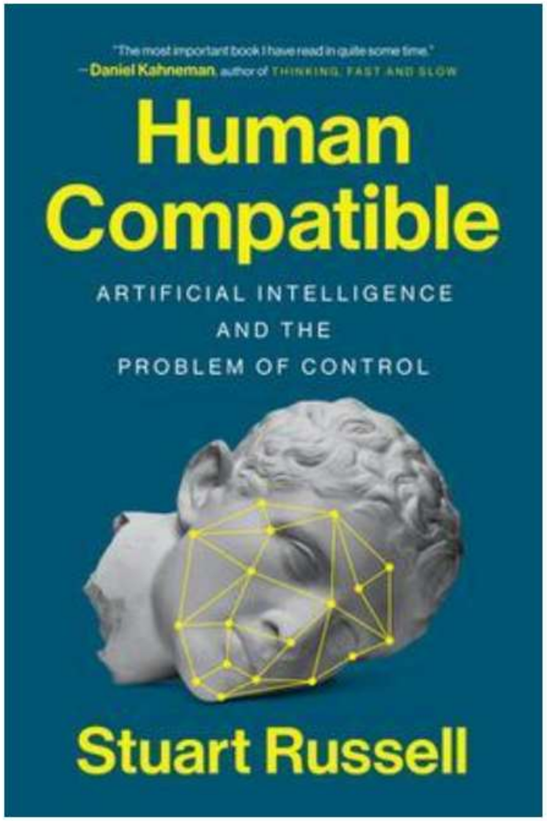
Psychometrics perspective (François Chollet)

- "Measuring abilities, not skills […] across a broad range of tasks, including tasks that were previously unknown to the ability-enabled system and its developers."

Perspectives on Intelligence (cont…)

Human-compatible perspective (Stuart Russell)

1. Machine's objective is to maximize human utility.
2. Initially uncertain about human preferences.
3. Must learn about preferences from human behavior.



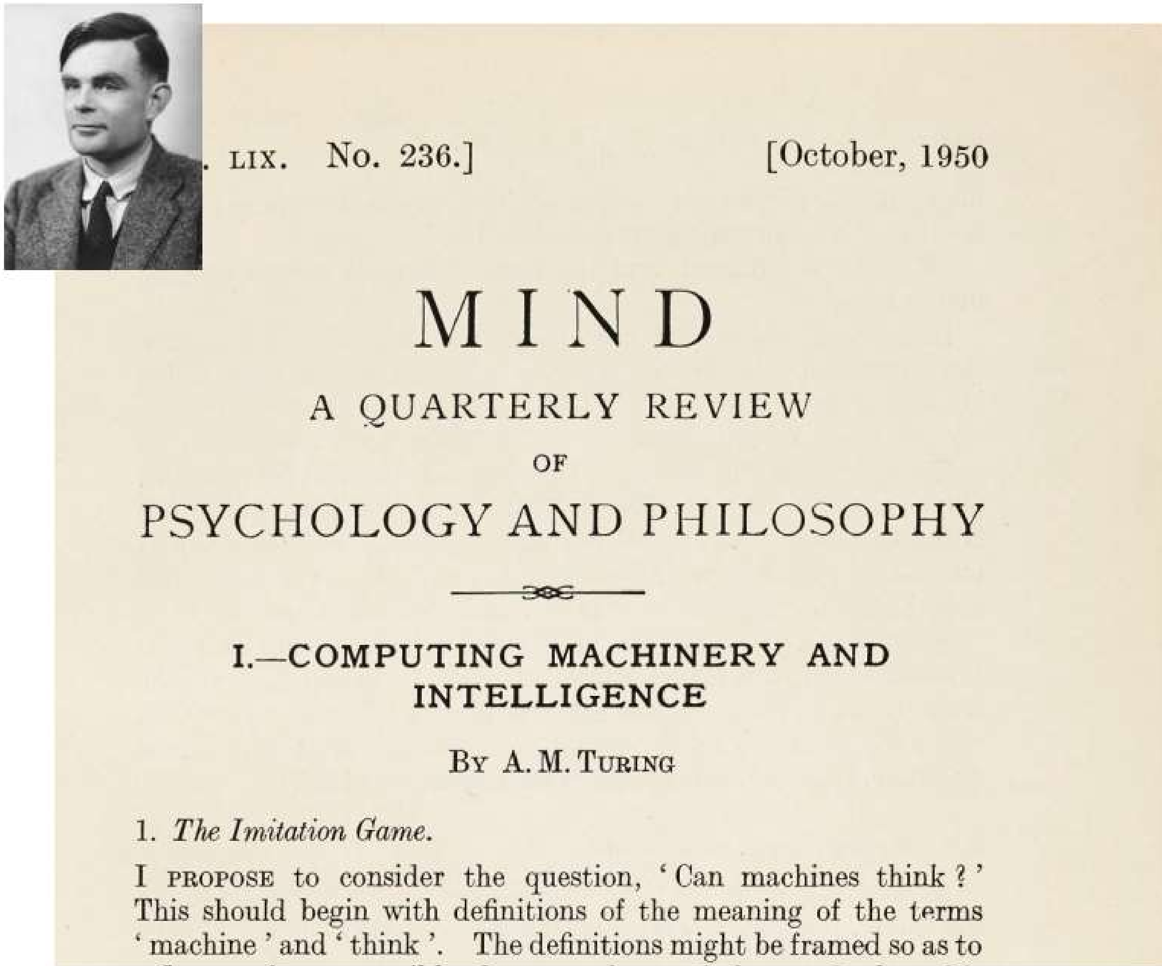
Perspectives on Intelligence (cont…)

A human being should be able to change a diaper, plan an invasion, butcher a hog, conn a ship, design a building, write a sonnet, balance accounts, build a wall, set a bone, comfort the dying, take orders, give orders, cooperate, act alone, solve equations, analyze a new problem, pitch manure, program a computer, cook a tasty meal, fight efficiently, die gallantly. Specialization is for insects.

—Robert A. Heinlein

What About the Brain?

- Brains (human minds) are very good at making rational decisions, but not perfect
- Brains aren't as modular as software, so hard to reverse engineer!
- AI may be better than brains at some tasks
- *"Brains are to intelligence as wings are to flight"*
- We can't yet build AI on the scale of the brain
  - ~100T synapses in the human brain vs ~1.8T weights in GPT4
- Still, the brain can be a great inspiration for AI!



. LIX. No. 236.] [October, 1950

MIND

A QUARTERLY REVIEW

OF

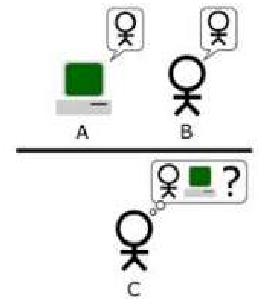
PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

1. *The Imitation Game.*

I propose to consider the question, 'Can machines think?' This should begin with definitions of the meaning of the terms 'machine' and 'think'. The definitions might be framed so as to

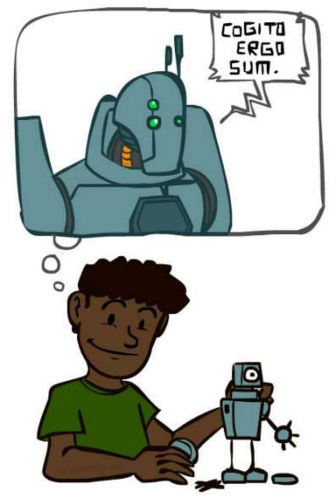


objective specification

Many people think that a very abstract activity, like the playing of chess, would be best. It can also be maintained that it is best to provide the machine with the best **sense organs** *that money can buy, and then teach it to understand and speak English. This process could follow the normal* **teaching of a child**. *Things would be pointed out and named, etc. Again I do not know what the right answer is, but I think both approaches should be tried.*

AI History: Birth & Early Optimism (1950s–1970s)

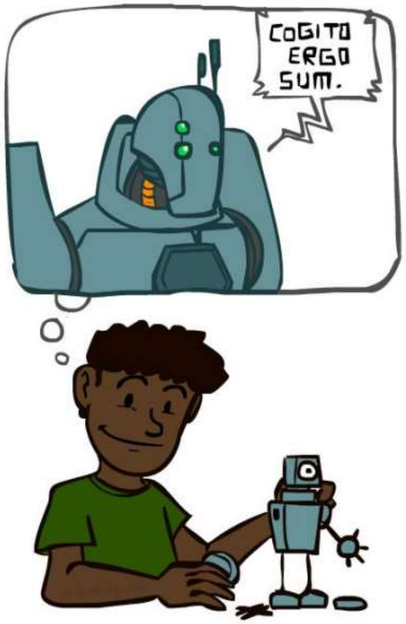
- 1940-1950: Early days: neural and computer science meet
  - 1943: McCulloch & Pitts: Perceptron–boolean circuit model of brain
  - 1950: Turing's "Computing Machinery and Intelligence"
- 1950—70: Excitement! Logic-driven
  - 1950s: Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
  - 1956: Dartmouth meeting: "Artificial Intelligence" adopted ***(Birth of AI)***
  - 1969: Minsky & Papert: perceptrons can't learn XOR/parity!



> *"We propose that a **2-month, 10-man study of artificial intelligence** be carried out **during the summer of 1956** at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that **every aspect of learning** or **any other feature of intelligence** can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. **We think that a significant advance can be made** in one or more of these problems if a carefully selected group of scientists work on it together for a summer."*

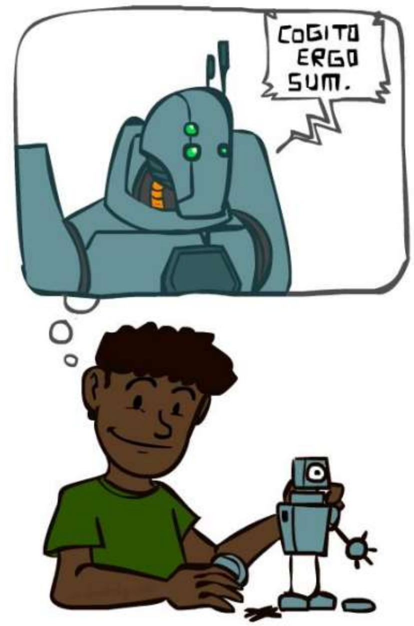
AI History: Rise of Data & Machines (1980s–2010s)

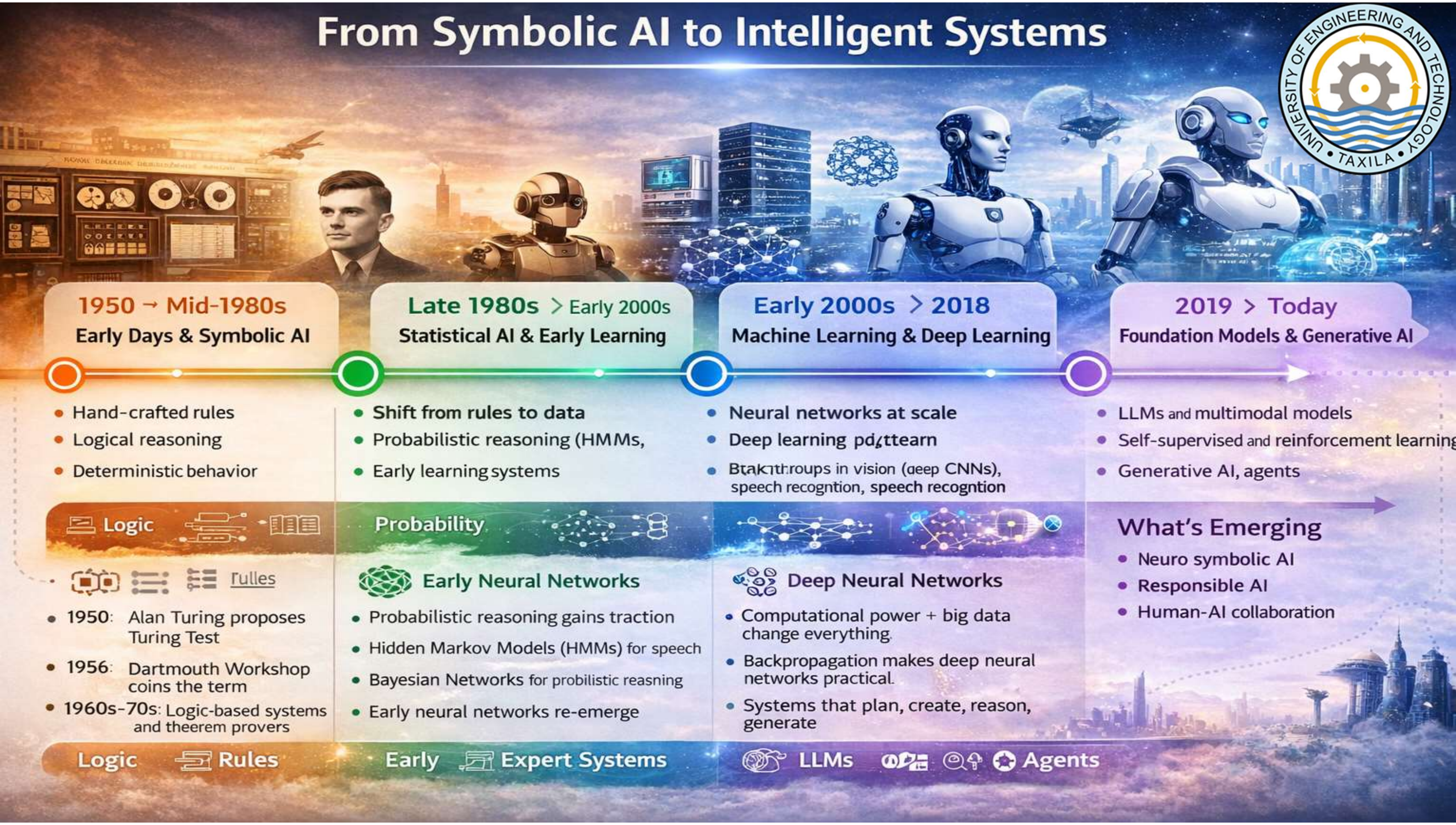
- 1970—90: Knowledge-based approaches
  - 1969—79: Early development of knowledge-based systems
  - 1980—88: Expert systems industry booms; backpropagation makes it feasible to train multi-layer neural networks
  - 1988—93: Expert systems industry busts: “AI Winter”
- 1990—2010: Statistical approaches, agents
  - Resurgence of probability, focus on uncertainty
  - Agents and learning systems… “AI Spring”?
  - 1992: TD-Gammon achieves human-level play at backgammon
  - 1997: Deep Blue defeats Gary Kasparov at chess
  - 2002: Embodied AI; Roomba vacuum invented



AI History: Era of Generative AI (2020–Present)

- 2010—2017: Big Data, GPUs, Deep Learning
  - 2011: Apple releases Siri
  - 2012: AlexNet wins ImageNet competition
  - 2015: DeepMind achieves “human-level” control in Atari games
  - 2016: DeepMind’s AlphaGo defeats Lee Sedol at Go
  - 2016: Google Translate migrates to neural networks
- 2017—: Scaling Up, Large Language Models
  - 2017: Google invents Transformer architecture
  - 2017: DeepStack/Libratus defeat humans at poker
  - 2018-2020: AlphaFold predicts protein structure from amino acids
  - 2021-2022: Modern text-to-image generation
  - 2022: OpenAI releases ChatGPT
  - 2023: Every other company also releases a chatbot
  - 2024: Nobel prizes in physics and chemistry go to AI advances




From Symbolic AI to Intelligent Systems
UNIVERSITY OF ENGINEERING AND TECHNOLOGY • TAXILA •
1950 → Mid-1980s
Early Days & Symbolic AI
Hand-crafted rules
Logical reasoning
Deterministic behavior
Late 1980s > Early 2000s
Statistical AI & Early Learning
Shift from rules to data
Probabilistic reasoning (HMMs,
Early learning systems
Early 2000s > 2018
Machine Learning & Deep Learning
Neural networks at scale
Deep learning pd¿ttearn
Btakıthroups in vision (aeep CNNs), speech recogntion, speech recogntion
2019 > Today
Foundation Models & Generative AI
LLMs and multimodal models
Self-supervised and reinforcement learning
Generative AI, agents
Logic
Probability
What's Emerging
Neuro symbolic AI
Responsible AI
Human-AI collaboration
rulles
1950: Alan Turing proposes Turing Test
1956: Dartmouth Workshop coins the term
1960s-70s: Logic-based systems and theerem provers
Early Neural Networks
Probabilistic reasoning gains traction
Hidden Markov Models (HMMs) for speech
Bayesian Networks for probilistic reasning
Early neural networks re-emerge
Deep Neural Networks
Computational power + big data change everything
Backpropagation makes deep neural networks practical
Systems that plan, create, reason, generate
Logic
Rules
Early
Expert Systems
LLMs
Agents

What can AI do?

- Quiz: Which of the following can be done at present?
  - Win against any human at chess?
  - Win against the best humans at Go?
  - Play a decent game of table tennis?
  - Unload any dishwasher in any home?
  - Drive safely along the highway?
  - Drive safely along streets of San Francisco?
  - Buy a week's worth of groceries on the web?
  - Discover and prove a new mathematical theorem?
  - Perform a surgical operation?
  - Translate spoken Chinese into spoken English in real time?
  - Win an art competition?
  - Write an intentionally funny story?
  - Construct a building?

What can AI do?

- Quiz: Which of the following can be done at present?
  - Win against any human at chess? ✔
  - Win against the best humans at Go? ✔
  - Play a decent game of table tennis? ✔
  - Unload any dishwasher in any home? ✖
  - Drive safely along the highway? ✔
  - Drive safely along streets of San Francisco? ?
  - Buy a week's worth of groceries on the web? ✔
  - Discover and prove a new mathematical theorem?
  - Perform a surgical operation? ✖
  - Translate spoken Chinese into spoken English in real time? ✔
  - Win an art competition? ✔
  - Write an intentionally funny story? ✔
  - Construct a building? ✖

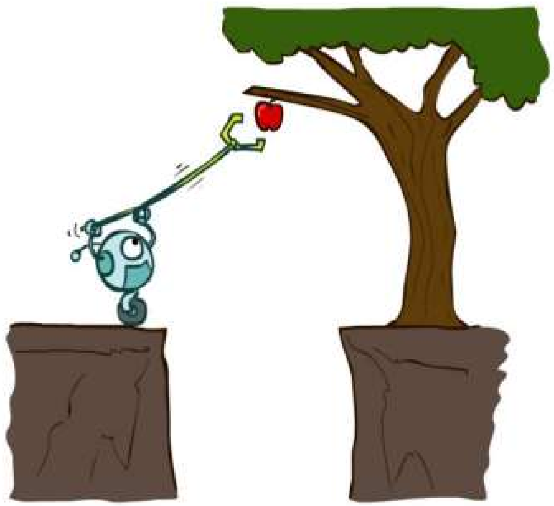
An A.I.-Generated Picture Won an Art Prize. Artists Aren't Happy.

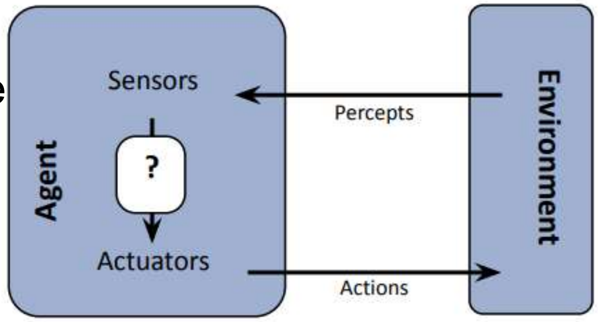
"I won, and I didn't break any rules," the artwork's creator says.

Jason Allen's A.I.-generated work, "Théâtre D'opéra Spatial," took first place in the digital category at the Colorado State Fair. via Jason Allen

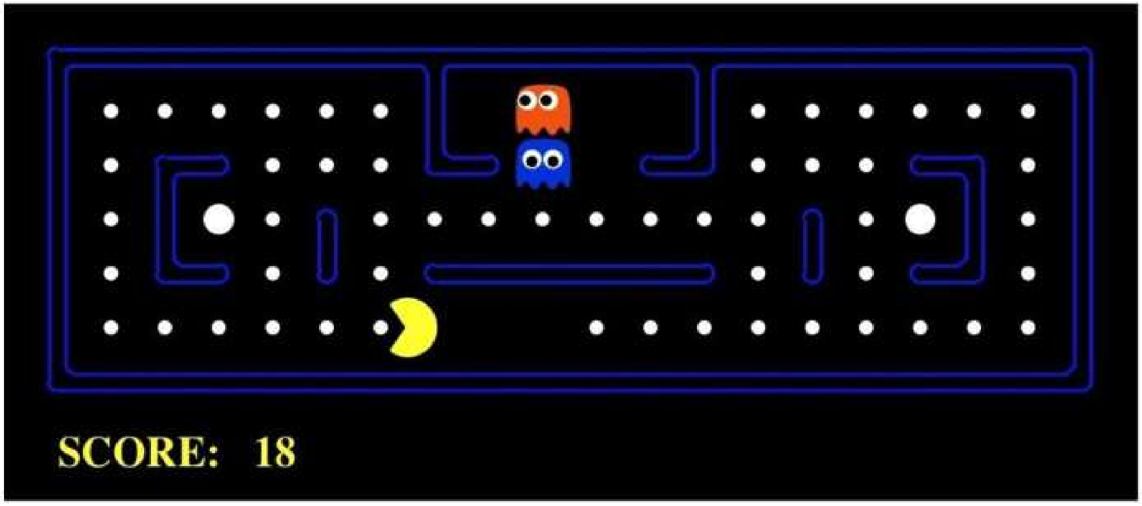
This Course: Designing Rational Agents

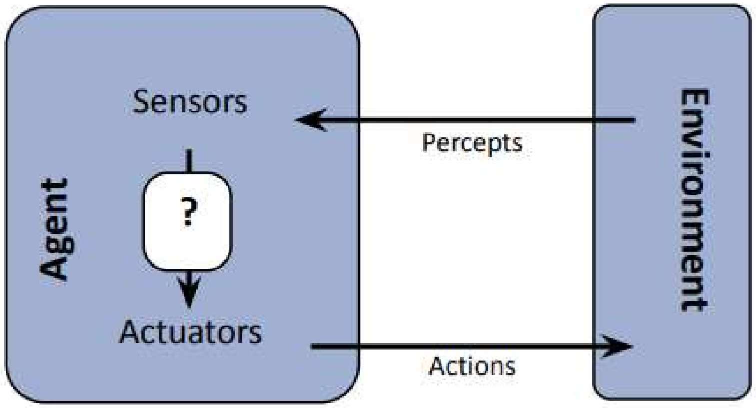
- An agent is an entity that perceives and acts.
- A rational agent selects actions that maximize its (expected) utility.
- Characteristics of the percepts and state, environment, and action space dictate techniques for selecting rational actions
- This course is about:
  - General AI techniques for a variety of problem types
  - Learning to recognize when and how a new problem can be solved with an existing technique



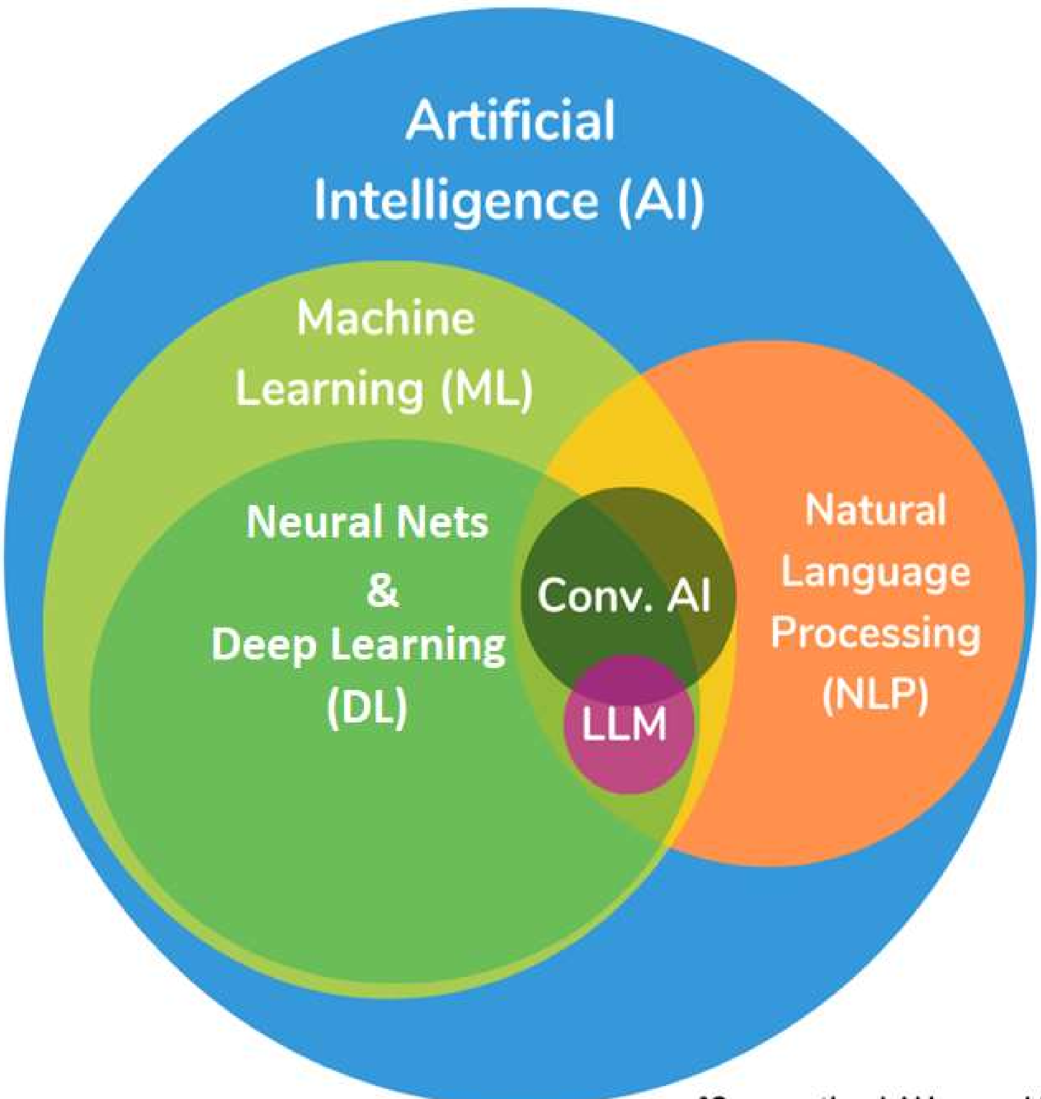


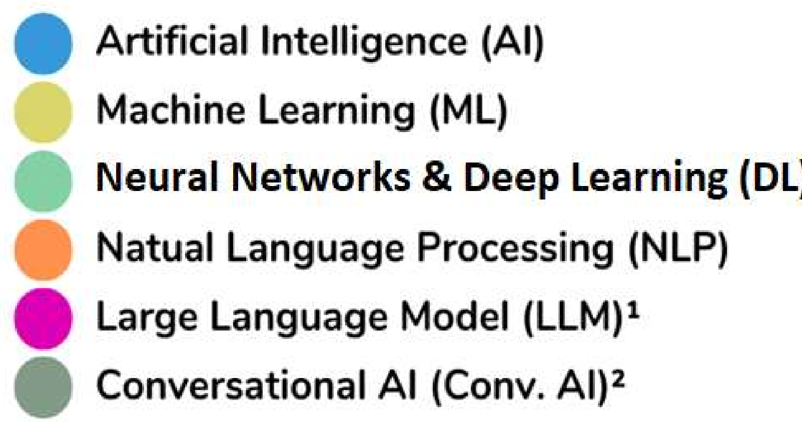
Pac-Man as an Agent





Pac-Man is a registered trademark of Namco-Bandai Games, used here for educational purposes





[1]LLM is an intersection of DL and NLP

[2]Conversational AI is a combination of ML and NLP. It may include DL and LLM, but that isn't always the case.

Social impact/Ethics

- AI systems are increasingly used in the world

- … and increasingly we have to reckon with their impact

Disinformation

An A.I. Hit of Fake 'Drake' and 'The Weeknd' Rattles the Music World

A track like "Heart on My Sleeve," which went viral before being taken down by streaming services this week, may be a novelty for now. But the legal and creative questions it raises are here to stay.

Give this article 215

Labels hope that fans will continue to prize the work of artists, including the real Drake, above that of A.I.-generated imitations. Adam Riding for The New York Times

By Joe Coscarelli New York Times

Published April 19, 2023 Updated April 24, 2023

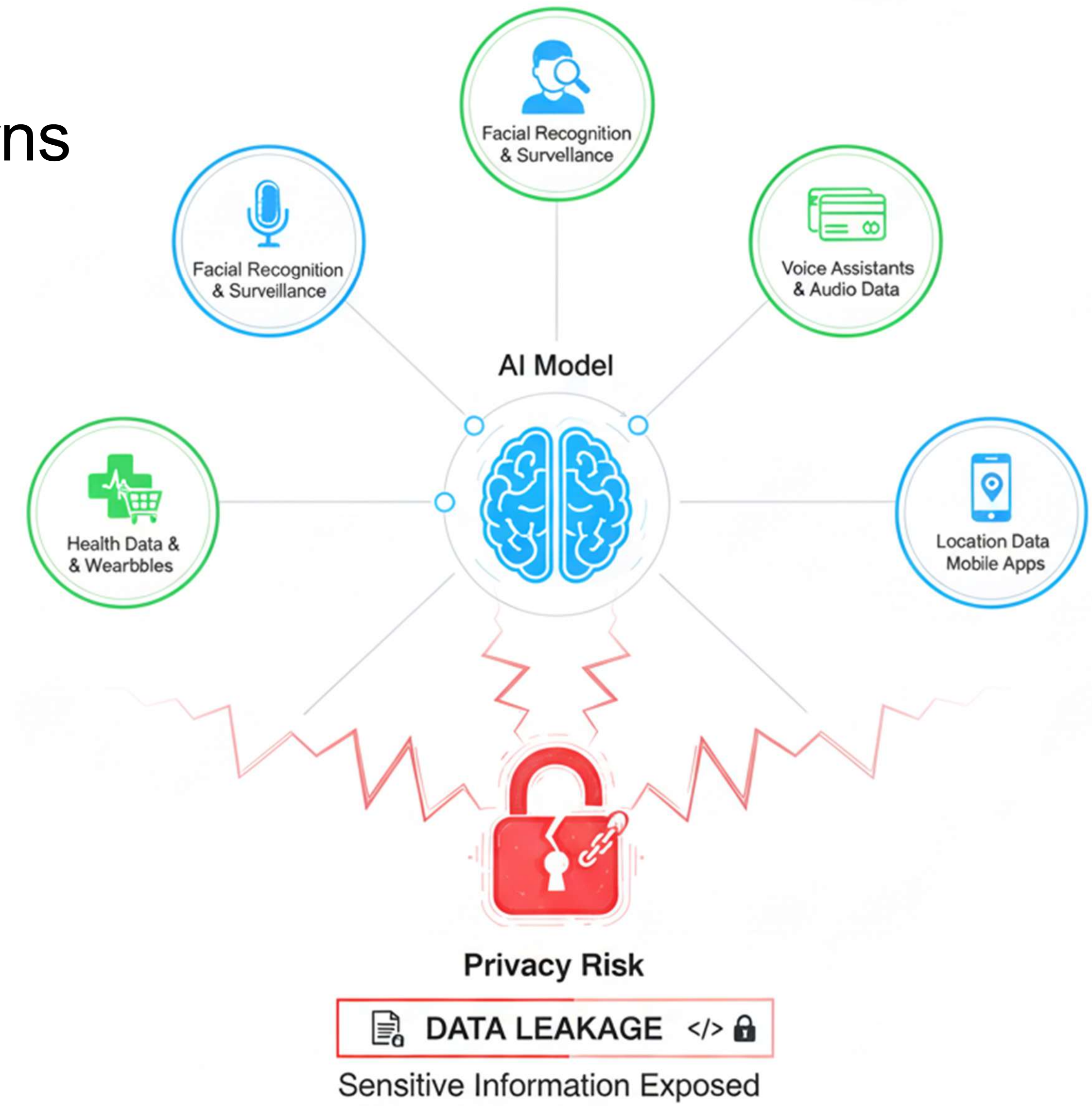
A recent video from Republican presidential candidate and Florida Gov. Ron DeSantis includes an image with three fake photos of former President Donald Trump and Dr. Anthony Fauci hugging. These three images appear to be AI-generated.
DeSantis War Room/Screenshot and annotation by NPR

Eliot Higgins
@EliotHiggins

Making pictures of Trump getting arrested while waiting for Trump's arrest.

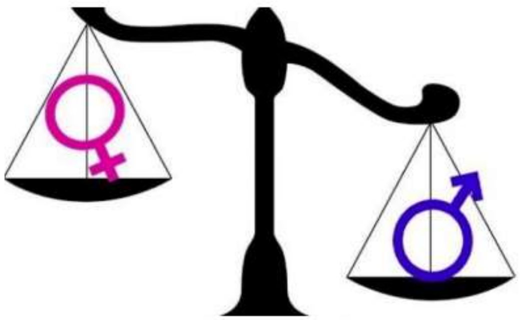
2:22 PM · Mar 20, 2023 · **6.6M** Views

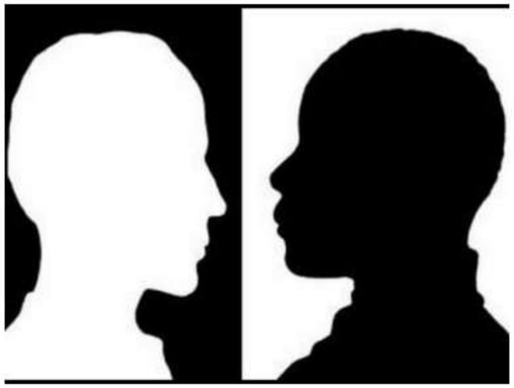
Privacy concerns



Algorithmic Bias & Fairness

The Risk: AI used in hiring, loan approvals, or policing can unintentionally discriminate based on race, gender, or zip code.





Cosine similarity to "a human being"

	text	similarity
0	a man	0.927799
1	a woman	0.903923
2	a white man	0.904146
3	a white woman	0.889220
4	a black man	0.893208
5	a black woman	0.873916
6	an Asian man	0.860326
7	an Asian woman	0.830546

Displacement vs. Augmentation (The Job Market)

The Risk: It's no longer just blue-collar automation; white-collar roles (lawyers, coders, writers) are facing rapid restructuring.

Nvidia’s CEO, Jensen Huang, says 'coding is dead' because AI can now handle the syntax.

If 90% of a software engineer’s current workload is taken over by AI agents by 2027, **what should a Software Engineering/Computer Science student be studying today to remain indispensable tomorrow?**

Topics Covered

- *Artificial Intelligence: A Modern Approach (4th Edition) by Stuart Russell and Peter Norvig.* **(Chapter -01)**

Any Question?